1 Entanglement entropies

Entanglement can be quantified in various different fashions. Here, we want to explore the von Neumann and the Rényi entropies.

1.1 The von Neumann entanglement entropy

The von Neumann entropy is defined as

$$S(\rho_A) \equiv -\operatorname{Tr}\big(\rho_A \ln \rho_A\big) \tag{1}$$

S is zero for two unentangled systems and non-zero for entangled systems.

Claim: $S(\rho_A) \equiv -\operatorname{Tr}\big(\rho_A \ln \rho_A\big) = -\sum_i \lambda_i \ln \lambda_i = -\sum_i s_i^2 \ln s_i^2$.

Proof: Since ρ_A is hermitian, we can use the spectral theorem to write it as $\rho_A = \sum_i \lambda_i P_i$ where λ_i are its eigenvalues and the eigenvectors $|u_i\rangle$ give $P_i = |u_i\rangle\langle u_i|$, since

$$\rho_A = UDU^\dagger = \big(|u_1\rangle, \ldots, |u_n\rangle\big) \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix} \begin{pmatrix} \langle u_1| \\ \vdots \\ \langle u_n| \end{pmatrix}, \qquad \langle u_i|u_j\rangle = \delta_{ij}. \tag{2}$$

For a function f defined on the spectrum of ρ_A, the spectral functional calculus[1] gives

$$f(\rho_A) \equiv \sum_i f(\lambda_i)\, |u_i\rangle\langle u_i| = U \operatorname{diag}\big(f(\lambda_i)\big) U^\dagger. \tag{3}$$

Therefore, one can rewrite the von Neumann entanglement entropy as

$$\begin{aligned} S(\rho_A) &\equiv -\operatorname{Tr}\big(\rho_A \ln \rho_A\big) && (4)\\ &= -\operatorname{Tr}\big(\rho_A\, U \ln(\operatorname{diag}(\lambda_i))\, U^\dagger\big) \\ &= -\operatorname{Tr}\big(U \operatorname{diag}(\lambda_i) \underbrace{U^\dagger U}_{=\mathbb{1}} \ln(\operatorname{diag}(\lambda_i)) U^\dagger\big) \\ &= -\sum_i \lambda_i \ln \lambda_i = -\sum_i s_i^2 \ln s_i^2. && (5) \end{aligned}$$

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Example

In our case, for example, $L = 2$, $N_f = 1$, $N_{\text{ph}} = 1$:

$$\begin{aligned} &\{|0\rangle, |1\rangle\} \;\&\; \{|01\rangle, |10\rangle\} \quad \text{as bases of } \mathcal{H}_b \;\&\; \mathcal{H}_f, && (6)\\ &\Rightarrow \{|0,01\rangle;\ |0,10\rangle;\ |1,01\rangle;\ |1,10\rangle\} \quad \text{as basis of } \mathcal{H}_b \otimes \mathcal{H}_f, && (7)\\ &\Rightarrow |\psi\rangle = \sum_{i=1}^{2}\sum_{j=1}^{2} \psi_{ij}\, |i\rangle_b\, |j\rangle_f \quad \text{with } (\psi)_{ij} = \psi_{ij} \;\&\; \psi \in \mathbb{C}^{2\times 2}, && (8)\\ &\Rightarrow S(\rho_A) = -\sum_\ell s_\ell^2 \ln s_\ell^2. && (9) \end{aligned}$$

[1] See for example Nicholas, J. H. (2008). *Functions of matrices: theory and computation.*

Note that since $\operatorname{tr}\rho^{AB} = \operatorname{tr}\rho^{A} = \operatorname{tr}\rho^{B} = 1$ we must have $\sum_\ell \lambda_\ell = \sum_\ell s_\ell^2 = 1$.

1.2 Direct computation using the SVD

The entanglement entropy for the ground state can be computed directly as follows:

1. Obtain an approximation of the ground state $|\psi_0\rangle = \sum_\alpha c_\alpha |\alpha\rangle \in \mathcal{H}^{AB} = \mathcal{H}^A \otimes \mathcal{H}^B$ as some parametrized variational state $\psi_\theta(\alpha)$.
2. Use the variational state (e.g. $\psi_\theta(\alpha) = c_\alpha$) to evaluate the amplitudes on every allowed configuration α and normalize the resulting coefficient vector.
3. Reshape the coefficients into the matrix $\psi \in \mathbb{C}^{d_A \times d_B}$, respecting the ordering of the product basis.
4. Compute the SVD of ψ to obtain the Schmidt values s_i, yielding the entropy $S = -\sum_i s_i^2 \ln s_i^2$

This approach is also straightforward with exact diagonalization since the ground state is represented by a vector consisting of the desired amplitudes. However, this quickly becomes unfeasible for ED and variational states as the Hilbert space grows exponentially with the system size.

1.3 Computation of the von Neumann entropy in VMC

1.3.1 Constructing the reduced density matrix

Given that $d_A \ll d_B$, one can estimate the small reduced density matrix ρ_A and diagonalize it directly. For illustrative purposes, let subsystem A describe a single photon mode and let subsystem B describe a fermionic system.

Let $n = 0, \ldots, N_{\text{ph}}$ denote the photon occupation and σ a fermionic occupation configuration. Define the wavefunction amplitudes by

$$\psi(n, \sigma) \equiv \langle n, \sigma | \psi \rangle \,. \tag{10}$$

For a normalized state, the matrix elements of ρ_A follow from **????**

$$(\rho_A)_{nn'} = \sum_\sigma \psi(n, \sigma)\psi^*(n', \sigma) \tag{11}$$

Since variational states do not necessarily provide normalized amplitudes, we might have to normalize them in the above formula. Defining

$$Z \equiv \sum_{m,\sigma} |\psi(m, \sigma)|^2, \tag{12}$$

we can write the normalized reduced density matrix more generally as

$$\boxed{(\rho_A)_{nn'} = \frac{\sum_\sigma \psi(n, \sigma)\psi^*(n', \sigma)}{Z}}. \tag{13}$$

For a normalized state, $Z = 1$.

1.3.2 A Monte Carlo estimator for the reduced density matrix

For each fermionic configuration σ, define

$$u_n(\sigma) \equiv \frac{\psi(n,\sigma)}{\sqrt{\sum_{m=0}^{N_{\rm ph}} |\psi(m,\sigma)|^2}}, \qquad \boldsymbol{u}(\sigma) = \begin{pmatrix} u_0(\sigma) \\ \vdots \\ u_{N_{\rm ph}}(\sigma) \end{pmatrix} \tag{14}$$

only if the denominator is non-zero. This is almost always the case when using Monte Carlo sampling since these configurations will never be visited, unless you start in one of them. This vector is normalized for every σ with non-zero marginal probability:

$$\boldsymbol{u}(\sigma)^\dagger \boldsymbol{u}(\sigma) = \sum_{n=0}^{N_{\rm ph}} |u_n(\sigma)|^2 = 1. \tag{15}$$

The matrix elements of its outer product are

$$\big(\boldsymbol{u}(\sigma)\boldsymbol{u}(\sigma)^\dagger\big)_{nn'} = \frac{\psi(n,\sigma)\psi^*(n',\sigma)}{\sum_{m=0}^{N_{\rm ph}} |\psi(m,\sigma)|^2}. \tag{16}$$

Now consider the marginal probability distribution of the fermionic configurations,

$$p(\sigma) = \sum_{m=0}^{N_{\rm ph}} p(m,\sigma) \tag{17}$$

$$\equiv \frac{\sum_{m=0}^{N_{\rm ph}} |\psi(m,\sigma)|^2}{\sum_{m,\sigma'} |\psi(m,\sigma')|^2}. \tag{18}$$

Taking the expectation value with respect to this distribution yields

$$\begin{aligned}
\mathbb{E}_{\sigma\sim p(\sigma)}\big[\boldsymbol{u}(\sigma)\boldsymbol{u}(\sigma)^\dagger\big]_{nn'} &= \sum_\sigma \frac{\psi(n,\sigma)\psi^*(n',\sigma)}{\sum_m |\psi(m,\sigma)|^2}\, p(\sigma) \\
&= \sum_\sigma \frac{\psi(n,\sigma)\psi^*(n',\sigma)}{\sum_m |\psi(m,\sigma)|^2}\, \frac{\sum_m |\psi(m,\sigma)|^2}{\sum_{m,\sigma'} |\psi(m,\sigma')|^2} \\
&= \frac{\sum_\sigma \psi(n,\sigma)\psi^*(n',\sigma)}{\sum_{m,\sigma'} |\psi(m,\sigma')|^2} \\
&= (\rho_A)_{nn'}.
\end{aligned} \tag{19}$$

Therefore,

$$\boxed{\rho_A = \mathbb{E}_{\sigma\sim p(\sigma)}\big[\boldsymbol{u}(\sigma)\boldsymbol{u}(\sigma)^\dagger\big]}. \tag{20}$$

Moreover, since we are only interested in an expectation value over σ, we have

$$\begin{aligned}
\mathbb{E}_{(n,\sigma)\sim p(n,\sigma)}\big[F(\sigma)\big] &= \sum_{n,\sigma} p(n,\sigma) F(\sigma) \\
&= \sum_\sigma p(\sigma) F(\sigma) \\
&= \mathbb{E}_{\sigma\sim p(\sigma)}\big[F(\sigma)\big].
\end{aligned} \tag{21}$$

Thus, we can sample either from $p(\sigma)$ or from $p(n,\sigma)$, depending on what is more convenient.

1.3.3 Computation using variational Monte Carlo

The resulting procedure is:

1. Sample a configuration $(\tilde{n}, \sigma)$ using the VMC sampler, i.e. from the joint distribution

$$p(n, \sigma) = \frac{|\psi(n, \sigma)|^2}{Z}. \tag{22}$$

   Discard $\tilde{n}$ and retain σ. Note, that Z does not need to be known in VMC. The retained configurations follow the marginal distribution $p(\sigma)$ (see eq. (21)).

2. Evaluate all amplitudes $\psi(n, \sigma)$ for $n = 0, \ldots, N_{\text{ph}}$ and construct $\boldsymbol{u}(\sigma)$.

3. Compute $\boldsymbol{u}(\sigma)\boldsymbol{u}(\sigma)^\dagger$. Repeat the preceding steps and average the resulting matrices:

$$\widehat{\rho}_A = \frac{1}{M} \sum_{k=1}^{M} \boldsymbol{u}(\sigma_k)\boldsymbol{u}(\sigma_k)^\dagger \approx \rho_A. \tag{23}$$

   The wide hat indicates that we only obtain an estimate of the true density matrix.

4. Diagonalize $\widehat{\rho}_A$ and use its eigenvalues $\widehat{\lambda}_i$ to estimate the von Neumann entropy:

$$\widehat{S} = -\sum_i \widehat{\lambda}_i \ln \widehat{\lambda}_i. \tag{24}$$

1.4 The Rényi entanglement entropies

The Rényi entropy of order n is defined as

$$S_n(\rho_A) \equiv \frac{1}{1-n} \ln(\text{tr}[\rho_A^n]) \tag{25}$$

$$= \frac{1}{1-n} \ln \left(\sum_{i=1}^{d_A} \lambda_i^n \right) \equiv S_n(\boldsymbol{\lambda}), \tag{26}$$

using again, that any hermitian matrix is diagonalizable and the cylic property of the trace. Keep in mind, that $0 \leq \lambda_i \leq 1$ and $\sum_i \lambda_i = 1$.

The Rényi entropy has the following useful properties, which we will prove subsequently:

- The Rényi entropy of order one reduces to the von Neumann entropy.
- $\lim_{n\to\infty} S_n(\boldsymbol{\lambda}) \equiv S_\infty(\boldsymbol{\lambda}) = -\ln \max_i \lambda_i$ and $S_2(\boldsymbol{\lambda}) \leq 2S_\infty(\boldsymbol{\lambda})$.
- We have $S_n \geq S_m$ for $n < m$ and therefore S_2 is a lower bound for the Von Neumann entropy S_1.

Claim: The Rényi entropy of order one reduces to the von Neumann entropy.

Proof: We can examine the first order by using l'Hôpital's rule, which yields

$$\begin{aligned} S_1(\rho_A) &= \lim_{n\to 1} \frac{\ln(\mathrm{tr}[\rho_A^n])}{1-n} \overset{\mathrm{H}}{=} \lim_{n\to 1} \frac{\left(\ln \sum_{i=1}^{d_A} \lambda_i^n\right)'}{(1-n)'} \\ &= -\lim_{n\to 1} \frac{\sum_{i=1}^{d_A} \lambda_i^n \ln \lambda_i}{\sum_{i=1}^{d_A} \lambda_i^n} = -\frac{\sum_{i=1}^{d_A} \lambda_i \ln \lambda_i}{\sum_{i=1}^{d_A} \lambda_i} \\ &= -\sum_{i=1}^{d_A} \lambda_i \ln \lambda_i. \end{aligned}$$

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Claim: $\lim_{n\to\infty} S_n(\boldsymbol{\lambda}) \equiv S_\infty(\boldsymbol{\lambda}) = -\ln \max_i \lambda_i$ and $S_2(\boldsymbol{\lambda}) \leq 2S_\infty(\boldsymbol{\lambda})$.

Proof: Order the different eigenvalues of ρ_A such that $\lambda_1 \leq \lambda_2 \leq \ldots \leq \lambda_2$. Furthermore, denote the number of degenerate $\lambda_1 \equiv \lambda_{\max}$ as m. It then follows

$$\begin{aligned} S_\infty(\boldsymbol{\lambda}) &= \lim_{n\to\infty} \frac{1}{1-n} \ln \left(\sum_{i=1}^{d_A} \lambda_i^n \right) \\ &= \lim_{n\to\infty} \frac{1}{1-n} \ln \left(\sum_{i=1}^{m} \lambda_{\max}^n + \sum_{i=m+1}^{d_A} \lambda_i^n \right) \\ &= \lim_{n\to\infty} \frac{1}{1-n} \ln \left(m\lambda_{\max}^n + 0 \right) \\ &= \lim_{n\to\infty} \frac{\ln m + n \ln \lambda_{\max}}{1-n} = -\ln \max_i \lambda_i. \end{aligned}$$

The second inequality quickly follows

$$S_2(\boldsymbol{\lambda}) = -\ln(\sum_{i=1}^{d_A} \lambda_i^n) \leq -\ln(\lambda_{\max}^2) = -2S_\infty(\boldsymbol{\lambda}),$$

since the logarithm decreases monotonically. ■

Claim: $S_n \geq S_m$ for $n < m$.

Proof: Define $z_i(n) = \lambda_i^n / \sum_{j=1}^{d_A} \lambda_j^n$, which is a probability distribution, and $F(n) = \ln \sum_{j=1}^{d_A} \lambda_j^n$. We take n to be a continuous variable and differentiate F with respect to it using again that $\frac{d}{dn}\lambda^n = \lambda^n \ln \lambda$:

$$\begin{aligned} F'(n) &= \frac{\sum_{i=1}^{d_A} \lambda_i^n \ln \lambda_i}{\sum_{j=1}^{d_A} \lambda_j^n} = \sum_i z_i(n) \ln \lambda_i \qquad \text{and} \\ F''(n) &= \sum_i z_i(n) (\ln \lambda_i)^2 - \left(\sum_i z_i(n) \ln \lambda_i \right)^2 \\ &= \mathrm{Var}_{z(n)}(\ln \lambda_i) \geq 0, \end{aligned}$$

which implies, that F' is non-decreasing. Thus, we can follow w.l.o.g for $x \geq n$

$$F(x) - F(n) = \int_n^x F'(s)ds \leq \int_n^x F'(n)ds = F'(n)(x-n),$$

which is equivalent to saying that F lays above all of its tangent, since it is convex. Since $F(1) = \ln \sum_i \lambda_i = \ln 1 = 0$, it follows that

$$F(1) = 0 \leq F(n) + F'(n)(1-n).$$

We can employ this now to yield

$$\frac{dS(n)}{dn} = \frac{d}{dn}\frac{F(n)}{1-n} = \frac{(1-n)F'(n) + F(n)}{(1-n)^2} \leq 0,$$

which proves, that S_n is non-increasing for $n \neq 1$. Finally, employing l'Hôpital's rule yields for $n = 1$ that

$$\lim_{n\to 1} = -\frac{1}{2}F''(1) \leq 0.$$

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1.5 Computation of the second order Rényi entropy in VMC

All orders of the Rényi entropies can also be exactly determined, if the reduced density matrix of one subpartition is given, by computing its eigenvalues and using eq. (26). The paper *Measuring Renyi Entanglement Entropy in Quantum Monte Carlo Simulations* by Hastings et al. shows how to compute the second order Rényi entropy and we outline their approach below.

The general idea is as follows. One defines an operator, where it can be shown that its expectation value is equal to the trace of the reduced density matrix squared. Then one only needs to show how this operator can actually be measured within VMC.

Let the full Hilbert space be partioned into A, B, such that

$$|\psi\rangle = \sum_{\alpha,\beta} C_{\alpha,\beta} |\alpha\rangle |\beta\rangle$$

and define

$$\begin{aligned} \mathrm{Swap}_A |\psi\rangle \otimes |\psi\rangle &= \mathrm{Swap}_A \sum_{\alpha_1,\beta_1,\alpha_2,\beta_2} C_{\alpha_1,\beta_1} D_{\alpha_2,\beta_2} |\alpha_1\rangle |\beta_1\rangle |\alpha_2\rangle |\beta_2\rangle \\ &\equiv \sum_{\alpha_1,\beta_1,\alpha_2,\beta_2} C_{\alpha_2,\beta_1} D_{\alpha_1,\beta_2} |\alpha_1\rangle |\beta_1\rangle |\alpha_2\rangle |\beta_2\rangle . \end{aligned} \tag{27}$$

The expectation value of this swap operator is given as

$$\begin{aligned}
\langle \mathrm{Swap}_A \rangle &\equiv \langle\psi| \langle\psi| \mathrm{Swap}_A |\psi\rangle |\psi\rangle \\
&= \sum_{\boldsymbol{\alpha},\boldsymbol{\beta},\boldsymbol{\alpha}',\boldsymbol{\beta}'} C^*_{\alpha_1,\beta_1} D^*_{\alpha_2,\beta_2} C_{\alpha_1',\beta_1'} D_{\alpha_2',\beta_2'} \langle\alpha_1\beta_1\alpha_2\beta_2| \mathrm{Swap}_A |\alpha_1'\beta_1'\alpha_2'\beta_2'\rangle \\
&= \sum_{\boldsymbol{\alpha},\boldsymbol{\beta},\boldsymbol{\alpha}',\boldsymbol{\beta}'} C^*_{\alpha_1,\beta_1} D^*_{\alpha_2,\beta_2} C_{\alpha_1',\beta_1'} D_{\alpha_2',\beta_2'} \delta_{\beta_1,\beta_1'}\delta_{\beta_2,\beta_2'}\delta_{\alpha_1,\alpha_2'}\delta_{\alpha_1,\alpha_1'} \\
&= \sum_{\boldsymbol{\alpha},\boldsymbol{\beta}} C_{\alpha_1,\beta_1} C^*_{\alpha_2,\beta_1} D^*_{\alpha_2,\beta_2} D_{\alpha_1,\beta_2}.
\end{aligned}$$

On the other side,

$$\begin{aligned}
\rho_A = \mathrm{tr}_B(\rho) &= \sum_{\boldsymbol{\alpha},\boldsymbol{\beta},\tilde{\beta}} \left\langle \tilde{\beta} \right| C_{\alpha_1,\beta_1} |\alpha_1\beta_1\rangle \langle\alpha_2,\beta_2| C^*_{\alpha_2,\beta_2} \left| \tilde{\beta} \right\rangle \\
&= \sum_{\beta,\alpha,\alpha'} C_{\alpha\beta} C^*_{\alpha'\beta} |\alpha\rangle \langle\alpha'| \qquad \text{such that} \\
(\rho_A)_{\alpha_1,\alpha_2} &= \sum_{\beta} C_{\alpha_1\beta} C^*_{\alpha_2\beta}.
\end{aligned}$$

Thus, we find that

$$\begin{aligned}
\langle \mathrm{Swap}_A \rangle &= \sum_{\alpha_1} \alpha_2 (\rho_A)_{\alpha_1,\alpha_2} (\rho_A)_{\alpha_2,\alpha_1} \\
&= \sum_{\alpha_1} \alpha_2 (\rho_A^2)_{\alpha_1,\alpha_1} = \mathrm{tr}(\rho_A^2).
\end{aligned} \tag{28}$$

Now how does this expectation value look like within VMC? Remember that

$$\begin{aligned}
\langle O \rangle &= \frac{\sum_{\boldsymbol{x}} |\psi(\boldsymbol{x})|^2 \sum_{\boldsymbol{x}'} \langle\boldsymbol{x}| O |\boldsymbol{x}'\rangle \frac{\langle\boldsymbol{x}'|\psi\rangle}{\langle\boldsymbol{x}|\psi\rangle}}{\sum_{\boldsymbol{x}''} |\psi(\boldsymbol{x}'')|^2} \\
&= \sum_{\boldsymbol{x}} p(\boldsymbol{x}) \sum_{\boldsymbol{x}'} \langle\boldsymbol{x}| O |\boldsymbol{x}'\rangle \frac{\langle\boldsymbol{x}'|\psi\rangle}{\langle\boldsymbol{x}|\psi\rangle}.
\end{aligned}$$

Computing this for $O = \mathrm{Swap}_A$, where $|\boldsymbol{x}\rangle \to |\alpha_1\beta_1\alpha_2\beta_2\rangle$ and $|\psi\rangle \to |\psi\rangle \otimes |\psi\rangle$ yields

$$\langle \mathrm{Swap}_A \rangle = \sum_{\boldsymbol{\alpha}\boldsymbol{\beta}} p(\alpha_1,\beta_1,\alpha_2,\beta_2) \frac{\psi(\alpha_1,\beta_2)\psi(\alpha_2,\beta_1)}{\psi(\alpha_1,\beta_1)\psi(\alpha_2,\beta_2)} \tag{29}$$

$$= \left\langle \frac{\psi(\alpha,\beta')\psi(\alpha',\beta)}{\psi(\alpha,\beta)\psi(\alpha',\beta')} \right\rangle, \tag{30}$$

with

$$p(\alpha_1,\beta_1,\alpha_2,\beta_2) = \frac{|\psi(\alpha_1,\beta_1)|^2 |\psi(\alpha_2,\beta_2)|^2}{\sum_{\boldsymbol{\alpha}'\boldsymbol{\beta}'} |\psi(\alpha_1',\beta_1')|^2 |\psi(\alpha_2',\beta_2')|^2}.$$

Thus, we can easily access this within VMC by sampling from the distribution given by $\psi(\alpha,\beta)\psi(\alpha',\beta')$ (see also NetKet documentation).

1.6 References

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